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RESEARCH ARTICLE

AIvaluate: A Multi-Agent Framework for Automated Answer Script Evaluation Using Large Language Models and Semantic Vector Indexing

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ABSTRACT In conventional educational environments, it is labor-intensive, subjective, and susceptible to human error to hand-mark descriptive answers. This article introduces AIvaluate, a modular, rubric-guided multi-agent system for theory-based student answer script marking automation. The system combines Optical Character Recognition (OCR) using Azure Document Intelligence with a Cleaning Agent powered by a Large Language Model (LLM) hosted on Azure AI Foundry that cleans the text obtained without losing semantic information and ensures anonymization. Answers are parsed using RegEx to de-sentence question identifiers and mapped into corresponding correct answers and faculty-defined marking schemes in a MongoDB database. The system supports structured input prompts, where rubrics are dynamically injected into the evaluation context for consistency and flexibility across subjects. An Evaluation Agent, also driven by the LLM, matches the sanitized student answer to the correct answer and provides both a score and one-line feedback. To supplement this generative scoring, a semantic similarity-based scoring is employed via FAISS and the all-MiniLM-L6-v2 transformer model; however, this vector-based pipeline is used purely for research benchmarking and does not contribute to the final marks. Experimental outcomes indicate strong correlation between AI-computed scores and human faculty ratings, particularly in the LLM-based pipeline. This system is designed particularly for technical and theory-based academic tests, and not for literary or creative questions where language usage is the prevailing criterion. The suggested approach reduces evaluation time by approximately 70–80% compared to manual correction, without compromising fairness and transparency.

INDEX TERMS Subjective answer evaluation, rubric-guided large language models, educational assessment automation, multi-agent systems, fairness in AI-based grading.

I. INTRODUCTION

The manual procedure of assessing handwritten answer scripts is time-consuming, error-prone, and laden with subjective biases. As class sizes grow, educational institutions require faster and more consistent methods for grading descriptive answers. Recent advances in Large Language Models (LLMs) have made it possible to design systems that assist in automating such evaluation tasks in a structured and controlled manner.

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This article presents AIvaluate, a modular, multi-agent system developed specifically for automating the evaluation of theory-based academic assessments. The system is organized into functional components, including pre-processing for text cleaning and anonymization, rubric management for handling faculty-defined grading schemes, and LLM-based scoring for answer evaluation. AIvaluate operates by taking student answers in plain text format, sanitizing them to remove personal identifiers, and comparing the content against rubrics provided by faculty. For each answer, the system refers to the assigned rubrics to determine a score and generate brief feedback. The rubrics can be updated by faculty according

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to subject requirements, ensuring flexibility without affecting the overall scoring method.

For benchmarking purposes, a semantic similarity model using FAISS and sentence-transformer embeddings is also applied. This allows comparison between vector-based scoring methods and rubric-guided LLM outputs. However, AIvaluate's primary scoring process is based solely on the language model guided by structured rubrics.

The system is applicable to examinations that involve structured, descriptive text responses. It is not designed to handle mathematical expressions, diagrammatic content, or creative writing tasks, where stylistic elements and complex symbols play a significant role.

The key contributions of this work are:

- Development of a modular evaluation framework combining pre-processing, rubric management, and LLM-based scoring agents.
- Implementation of a dynamic rubric-injection mechanism for prompt customization across subjects.
- Benchmarking rubric-guided LLM scoring against semantic similarity models to validate reliability.
- Incorporation of anonymization and fallback handling mechanisms to ensure fairness, transparency, and reviewability in automated grading.

II. RELATED WORK

In recent years, automated assessment systems have evolved significantly, particularly for evaluating descriptive and subjective responses. This section provides a detailed review of the existing literature related to this work. It covers the key technologies involved such as automated grading systems, Large Language Models (LLMs), semantic similarity methods like FAISS, optical character recognition (OCR), text pre-processing techniques, and hybrid grading approaches. All the references used here are directly cited from the bibliography section included in this paper.

A. AUTOMATED GRADING SYSTEMS FOR SUBJECTIVE ANSWERS

The first grading systems were rule-based and only considered the occurrence of predefined keywords and straightforward word-level similarities [2], [4]. They were unable to focus on matching semantic intent, since students provided their answers in different styles and formats. There were more sophisticated methods of grading that were developed beyond merely using keywords as evident in GRAD-AI [3] and the hybrid mark management system [4] to employ clustering, abstract syntax trees, and graphical flow representations. DAES [1] implemented topic modeling and semantic insight with T5 transformers, which increased the accuracy of prediction of scores. In addition, other authors have expanded the scope to include grading an open-book examination [6] and devised a method for cognitive assessment with multi-agent architectures [7].

These automated grading systems have taken some steps forward; however, they are mostly reliant on fixed models from ML and do not have use-based flexibility guided by rubrics or that facilitate modular separation of agents that was implemented earlier in AIvaluate.

B. LARGE LANGUAGE MODELS IN EDUCATIONAL NLP

Recently, LLM-based grading has been gaining in acceptance [8], [9], [10], [11], [12]. These models are closely correlated to human grading, particularly for descriptive responses, and backed by intraclass correlation values [10], [12]. Their zero-shot and few-shot capabilities allow them to be flexible for different question types. Other research has examined LLM usage for generating revision feedback and scoring([9], [13]).

However, most of these systems used LLMs as independent scorers, without using rubric injection or fallback management. AIvaluate fills this gap with dynamic rubric response and multi-agent reliability.

C. SEMANTIC SIMILARITY AND VECTOR SEARCH FOR AUTOMATED EVALUATION

Semantic similarity methods are prescribed as well-understood embedding models like Sentence-BERT and the Fast Approximate Nearest Neighbor algorithm (FAISS) [1], [2], [7], [14], [15]. The combination of TF-IDF and clustering has been explored in GRAD-AI [3] and other methods have transitioned to LDA or transformer embedding [1]. FAISS is still the preferred indexer for large volume similarity searching due to its speed and efficiency.

In AIvaluate, vector similarity with FAISS is utilized as a reference metric instead of the primary scoring method, though it does persist consistently as a score metric. The favor of rubric guided LLM scoring over ML methods in AIvaluate is for increased interpretability and flexibility.

D. OCR AND PREPROCESSING OF HANDWRITTEN SCRIPTS

Problems with handwritten script digitization are well known [16]. OCR tools have improved with AI-based correction models, such as Azure Document Intelligence [17], [18], using models based on transformer architectures and edge detection techniques [4]. There are even some systems that utilize vector-based NLP models after the OCR step to grade handwritten responses [19].

AIvaluate is designed to use text-based inputs, eliminating the Oxford OCR lever. This design choice limits the pathway for errors and the need for digitization during deployment focused on evaluation logic.

E. HYBRID SYSTEMS, EXPLAINABILITY, AND RESEARCH GAPS

Studies emphasize ongoing challenges related to explainability, fairness, and educator engagement in automated marking [13], [14]. Hybrid systems that combine fuzzy logic

and LLMs [20], or originality checking in two phases [18], have also been suggested. Domain-specific tools, such as those for predicting maths or gradings [17], are also available.

AIvaluate also tackles these challenges by proposing a modular, agent-based architecture (or architecture with agent-based components), with each functional element being separable, traceable and customizable through rubric-injection. This facilitates meaningful explanatory outputs for clarity, human engagement for oversight, and real-time adaptability for any subject.

III. SYSTEM ARCHITECTURE AND METHODOLOGY

This section will outline how the new AIvaluate system is designed and functions. The system is tailored for evaluating handwritten responses which are theory-related based on scanned exam papers. It integrates Optical Character Recognition (OCR), a Large Language Model (LLM) based pre-processing module, rubric-linked evaluation logic and provides a database of structured answers for automated grading and brief feedback.

The system design is modular and stagewise, allowing it to be scalable, manageable, and flexible to adapt to multiple subjects or exam formats for evaluation. The major design consideration eliminates the reliance on vector similarity scoring like FAISS and has much more controlled, rubric driven evaluation using generative reasoning from GPT-4o-mini.

A. OVERVIEW OF THE AIVALUATE PIPELINE

AIvaluate functions as a multi-stage pipeline of Azure-hosted micro-agents. The first phase begins with uploading a scanned handwritten answer script in PDF to the Alvaluate Frontend. At this point the PDF is forwarded to Azure Document Intelligence, performing Optical Character Recognition (OCR), extracting text from the scanned answer script, and storing the original text in Redis Cache DB as temporary in-memory storage.

Next, the Pre-processing AI agent (also powered by GPT-4o-mini hosted on Azure AI Foundry) cleans the text. Spelling mistakes are removed in the cleaning phase as are spaces and formatting, and student identifiers are anonymized. The normalized cleaned response is stored back to Redis in preparation for the next step.

Once the agent has completed the cleaning, regular expressions are used to parse through the cleaned content, and assign question numbers in a consistent format (e.g., Q1), Q2)). These question identification numbers associated with the question IDs are mapped to the corresponding entry in the Answer Key DB. The Answer Key DB is a MongoDB store that contains the question text, reference answers, rubrics for assessing, and total marks to allocate.

The Evaluating AI Agent (GPT-4o-mini) receives the cleaned student answer, reference answer, rubric details and maximum marks. The prompt is rubric-injected, and describes how the model is to be evaluated on accuracy, relevance, coverage of key points, and clarity. The Evaluating Agent returns a score and short one-line feedback.

Finally, the frontend aggregates per question the evaluated marks and feedback; and outputs this to the user in a structured and student-friendly format.

A high-level representation of the system is shown in Fig. 1.

B. OPTICAL CHARACTER RECOGNITION USING AZURE DOCUMENT INTELLIGENCE

At the start of the pipeline, Azure Document Intelligence, formerly Form Recognizer, is used to perform high-fidelity OCR on the uploaded handwritten PDFs. This is set up in Read Mode which extracts the raw line-by-line text along with bounding box metadata (not used by this system). High compatibility across multiple writing styles and layouts is important. The output is stored temporarily in Redis for processing in subsequent AI.

C. PRE-PROCESSING AGENT FOR CLEANING AND ANONYMIZATION USING GPT-4O-MINI

To ensure evaluation happens on clean structured content the extracted OCR text is sent to the Pre-processing AI agent that runs GPT-4o-mini. This agent does the following:

- Fixes common OCR spelling and punctuation mistakes,
- Fixes sentence alignment and formatting and restores it where possible,
- Fixes all student names, roll numbers, or identifiable information.

This cleaning step kept the semantic meaning but structured the input correctly and anonymized it. The cleaned version will replace the raw text in Redis for continuity.

D. QUESTION PARSING AND ANSWER KEY RETRIEVAL

After it is cleaned, the system applies a regular expression **(e.g., Q[0-9]+\))** identify the boundaries of each answer in the text. These question numbers are then leveraged to query the Answer Key DB, which is a MongoDB instance containing:

- Faculty-defined reference answers,
- Rubrics or evaluation guidelines,
- Maximum marks per question.

This schema allows AIvaluate to flexibly handle various question paper designs and subject matter and only utilize what is relevant to a student's answer.

E. EVALUATION AGENT FOR MARKING AND FEEDBACK

The Evaluating AI Agent which employs GPT-4o-mini structure receives the following inputs:

- Cleaned student answer
- Reference answer
- Rubric/ marking scheme
- Maximum marks

This agent evaluates the response using a custom prompt template that directly injects the rubric, so that the LLM

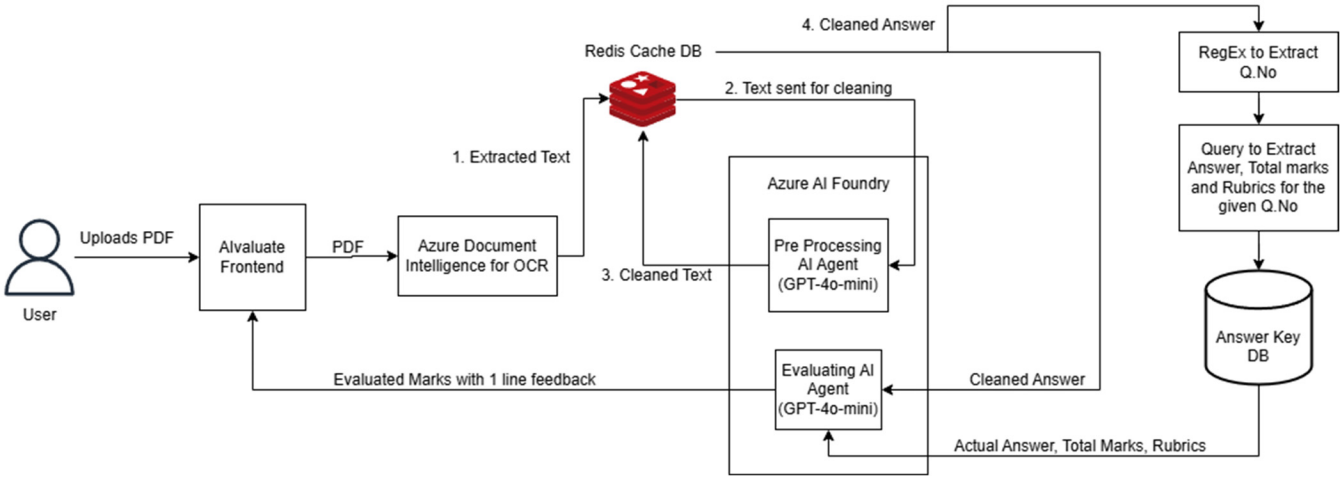


FIGURE 1. System architecture of AIvaluate.

can consider the rubric in context, and evaluate the response according to decisions provided by the context of the prompt. The agent:

- Evaluates correctness, coverage, and clarity as defined in the rubric
- Ignores cosmetic issues like grammar/spelling that were fixed during the Pre-processing Agent stage.
- Provides a score and one line of feedback explaining its conclusions.

Additionally, the system includes a retry and flagging mechanism to handle LLM inference failures gracefully. If the first evaluation attempt fails, a second attempt is made with adjusted parameters; persistent failures are flagged for manual review by faculty, ensuring robustness and operational reliability.

This architecture preserves consistency, fairness, and interpretability of marks awarded.

IV. EXPERIMENTAL SETUP AND DATASET DESCRIPTION

This section provides an overview of technical infrastructure, software tools, and evaluation methods that were used in the experiments. The focus of this research was to compare the scoring of an LLM-based evaluation system (AIvaluate) with a traditional semantic similarity method (FAISS) against a set of faculty-assigned gold standard marks.

A. ENVIRONMENTAL SETUP

The primary components of the system include:

1. **LLM Inference**: All GPT-based agents (Cleaning, Anonymization and Evaluation) were deployed on Azure AI Foundry, utilizing the **GPT-4o-mini** model for low-latency inference and cost efficiency.
2. **OCR Pipeline**: The OCR processing was handled via **Azure Document Intelligence**, which enables high-accuracy text extraction from scanned documents and handwritten answer scripts.
3. **Storage and Databases**:
   a. **Redis** was used as a fast-access in-memory store for intermediate results (raw and cleaned answers).
   b. **MongoDB** hosted the faculty-provided answer keys and associated metadata (question number, marks).
4. **Vector Similarity Infrastructure**: **FAISS** was implemented in a local Python environment for post-evaluation semantic similarity analysis. Sentence embeddings were computed using the **all-MiniLM-L6-v2** model from **SentenceTransformers**. FAISS was **not utilized in the core evaluation pipeline** but was retained to study alignment between LLM-based scores and embedding-based similarity.

All experiments were conducted on a machine with the following configuration:

1. **Processor**: AMD Ryzen 5 (4600H)
2. **RAM**: 16 GB
3. **GPU**: NVIDIA GTX 1650 (used for local model testing)
4. **Operating System**: Windows 11
5. **Python Version**: 3.10

B. EVALUATION METHODOLOGY

We conducted a comparative study between two evaluation agents:

1) LLM EVALUATION AGENT

Makes use of rubric-injected prompts in GPT-4o-mini to provide marks and give one-line feedback per answer.

2) FAISS-BASED EVALUATION

Along with the LLM-based Evaluation Agent, we investigate a second evaluation method using Facebook AI Similarity Search (FAISS) to assess the semantic similarity between student answers and a standard model answer. FAISS is the most commonly used library for performing similarity search

over high-dimensional vectors, especially for retrieving from large-scale embeddings of documents and text.

First, we embed each reference answer from the provided answer key (by the faculty), using the all-MiniLM-L6-v2 model from SentenceTransformers, which is most recognized for producing consistent sentence-level high-quality embeddings. These are used as the reference vectors for comparison.

The student answers, after the pre-processing with OCR and LLM, are transformed into their respective normalized embedding vectors. We perform similarity search from each student vector to each embedded reference answer vector, using the FAISS library. Although we are using a FAISS. IndexFlatIP index to calculate inner product similarity, this is similar to cosine similarity since our vectors have been L2-normalized.

The similarity score (ranging from 0 to 1) is then multiplied by the marked maximum for that question and exported to a single marking sheet. While it is unsupervised and without specific a rubric, it offers somewhat interpretable baseline measure of semantic nearness to ideal answers.

This FAISS-based evaluation method was included solely for research benchmarking, and all the students' scores were compared against the faculty scores with MAE, RMSE and Pearson Correlation Coefficient. The aim here is to measure how well it can locate similarity with human judgement and differentiate the relative accuracy of the LLM-based scoring model.

C. DATASET DESCRIPTION

The dataset comprises 110 answer scripts of students from three subjects: Software Engineering (39), Knowledge Engineering (63), and Cloud Computing (8). The documents were handwritten, scanned, and stored as PDFs. Each subject folder contains:

1. Answer_Scripts/<Subject>/ – Folder with student answer PDFs.
2. Answer_Keys/<Subject>.txt – Faculty-provided reference answers, structured in the format Q.No) <Answer> for easier parsing.
3. Cleaned/<Subject>/ – Folder storing cleaned answer scripts (as plain text), post OCR and AI-driven sanitization.

Each answer script contains multiple questions, with the expectation that every answer begins with a question number followed by a closing parenthesis (e.g., **Q1)**), ensuring robust RegEx-based parsing.

D. PREPROCESSING WORKFLOW

The pre-processing pipeline seeks to maintain semantic meaning while eliminating the noise from handwriting/scanning artifacts, OCR artifacts, and formatting inconsistency. It standardizes student writing without modifying their intended meaning, giving us a reliable source for objective evaluation later in the workflow. The workflow includes the following stages:

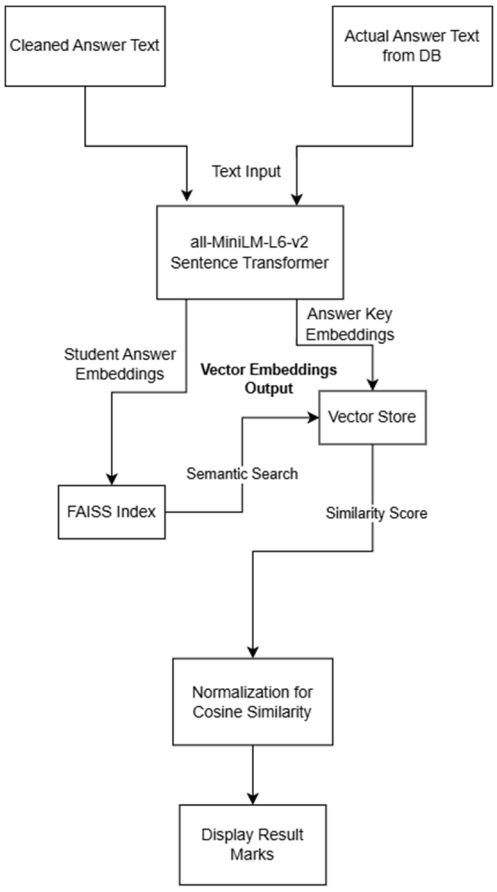


FIGURE 2. FAISS architecture diagram.

1. **Text extraction:** We input scanned PDF answer scripts into Azure Document Intelligence, which processes the scanned text and converts it to OCR and extracts it in a simple line-wise format.
2. **Cleaning:** Next, the raw OCR output is outputted to the Pre-processing Agent, which is powered by GPT-4o-mini, and performs spacing, punctuation and spelling corrections while maintaining the semantic structure of the answer.
3. **Redis Storage:** The original OCR's output and cleaned versions are stored in Redis for speedy retrieval for each evaluation agent.
4. **Parsing and Labelling:** Using a regular expression (Q\d+\)), we segment the cleaned text into answers. For each identified question, we match it up with the reference answer and the associated mark allocation stored in MongoDB.
5. **Evaluation phase:** At this stage of pre-processing, once parsing is complete, we send the cleaned student account and its reference to:
   a. the LLM-based Evaluation Agent (for rubric based scoring) and the FAISS
   b. based agent (for semantic similarity scoring - for research comparison).

The modularity of the architecture and workflow provides us with a controlled and repeatable method of evaluating student responses that is scalable across subjects and writing styles.

The dataset comprising student answer scripts and their respective answer keys, used for the experiments in this study, is publicly available at: https://github.com/Its-Suryakumar/AIvaluate-Dataset.

V. EVALUATION AND RESULTS

A. EVALUATION METRICS USED

1) MEAN ABSOLUTE ERROR

The MAE measures the average magnitude of the absolute errors between predicted scores and actual faculty-assigned scores.

$$MAE = \frac{1}{n}\sum_{i=1}^{n}|y_i - \hat{y}_i| \tag{1}$$

where:

i. y_i is the faculty-assigned mark for the ith sample.
ii. $\hat{y}_i$ is the AI-evaluated mark (LLM or FAISS).
iii. n is the number of answer scripts evaluated.

A lower MAE value indicates a closer agreement with human evaluations.

2) ROOT MEAN SQUARED ERROR (RMSE)

RMSE penalizes larger deviations more heavily than MAE and is sensitive to outliers.

$$\text{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2} \tag{2}$$

where:

1) y_i is the faculty-assigned mark for the ith sample.
2) $\hat{y}_i$ is the AI-evaluated mark (LLM or FAISS).

A lower RMSE indicates fewer high-magnitude errors in prediction.

3) PEARSON CORRELATION COEFFICIENT (PCC)

The PCC quantifies the linear correlation between AI-generated scores and faculty scores. It ranges from -1 (perfect negative correlation) to +1 (perfect positive correlation).

$$\text{PCC} = \frac{\sum_{i=1}^{n}(y_i - \bar{y})\left(\hat{y}_i - \bar{\hat{y}}\right)}{\sqrt{\sum_{i=1}^{n}(y_i - \bar{y})^2}\cdot\sqrt{\sum_{i=1}^{n}\left(\hat{y}_i - \bar{\hat{y}}\right)^2}} \tag{3}$$

where:

1) $\bar{y}$ is the mean of faculty marks
2) $\bar{\hat{y}}$ *is* the mean of AI-evaluated marks

A value close to $+1$ indicates strong agreement between the system and faculty grading patterns.

B. SUBJECT-WISE PERFORMANCE COMPARISON

1) SOFTWARE ENGINEERING (SE)

The Fig. 3 overlaid graph provides a visual comparison of the evaluation performance between the **LLM**, **FAISS**, and **Faculty** models for Software Engineering (SE). The graph highlights the correlation between predicted marks from both LLM and FAISS with the Faculty's marks across the subject.

Interpretation: The **SE FAISS** method shows a strong correlation with the faculty marks (0.91) and performs better in terms of MAE and RMSE compared to **SE LLM**. The **LLM** method, though showing a decent correlation of 0.66, is slightly less accurate in terms of MAE and RMSE.

TABLE 1. Evaluation metrics for Software Engineering (SE).

Model/Method	MAE	RMSE	Pearson Correlation
SE LLM	1.12	1.46	0.66
SE FAISS	1.55	1.98	0.91

TABLE 2. Evaluation metrics for Knowledge Engineering (KE).

Model/Method	MAE	RMSE	Pearson Correlation
KE LLM	0.88	1.06	0.80
KE FAISS	1.42	1.52	0.10

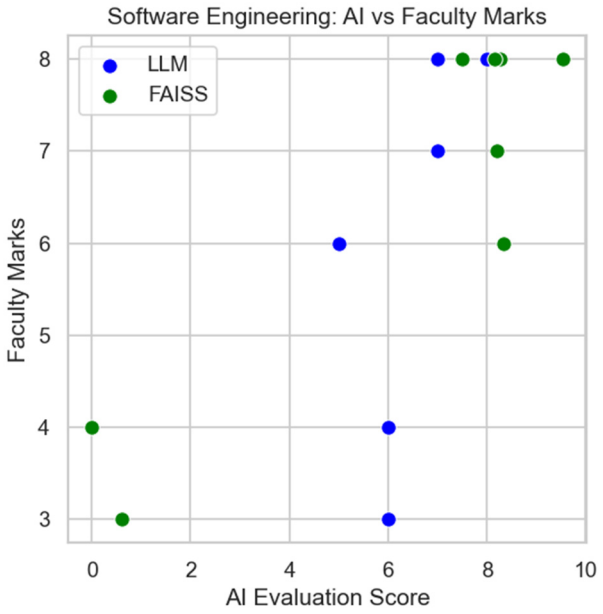


FIGURE 3. Overlaid graph of evaluation performance for Software Engineering.

2) KNOWLEDGE ENGINEERING (KE)

The Fig. 4 overlaid graph provides a visual comparison of the evaluation performance between the **LLM**, **FAISS**, and **Faculty** models for Knowledge Engineering (KE). The graph highlights the correlation between predicted marks from both LLM and FAISS with the Faculty's marks across the subject.

Interpretation: **KE LLM** outperforms **KE FAISS** with a lower MAE and RMSE, and shows a significantly better

TABLE 3. Evaluation metrics for Cloud Computing.

Model/Method	MAE	RMSE	Pearson Correlation
Cloud LLM	0.88	1.06	0.97
Cloud FAISS	1.54	1.89	0.95

Pearson correlation of 0.80. **KE FAISS** performs poorly with a weak correlation (0.10) and higher error metrics.

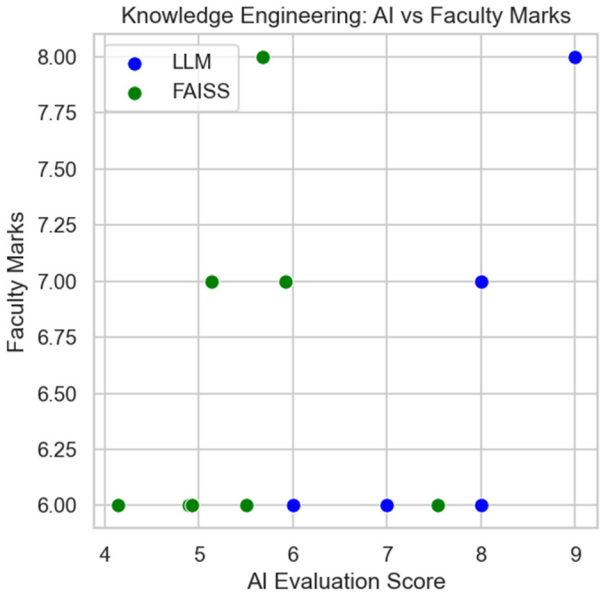


FIGURE 4. Overlaid graph of evaluation performance for Knowledge Engineering.

3) CLOUD COMPUTING

The Fig. 5 overlaid graph provides a visual comparison of the evaluation performance between the **LLM**, **FAISS**, and **Faculty** models for Cloud Computing. The graph highlights the correlation between predicted marks from both LLM and FAISS with the Faculty's marks across the subject.

Interpretation: **Cloud LLM** shows the best performance with the lowest MAE and RMSE, alongside an excellent Pearson correlation of 0.97. Although the Cloud-based FAISS model shows a slightly lower correlation score of 0.95, it still demonstrates strong performance relative to the other evaluated models.

C. CORRELATION MATRIX

We looked at how closely the scores from the **LLM and FAISS** matched those given by the faculty. For this analysis, we used **correlation values**. A high positive value generally means the methods gave similar results, while a value closer to zero means there wasn't much agreement between them. Negative values suggest that the methods went in different directions. The chart below clearly shows how each method

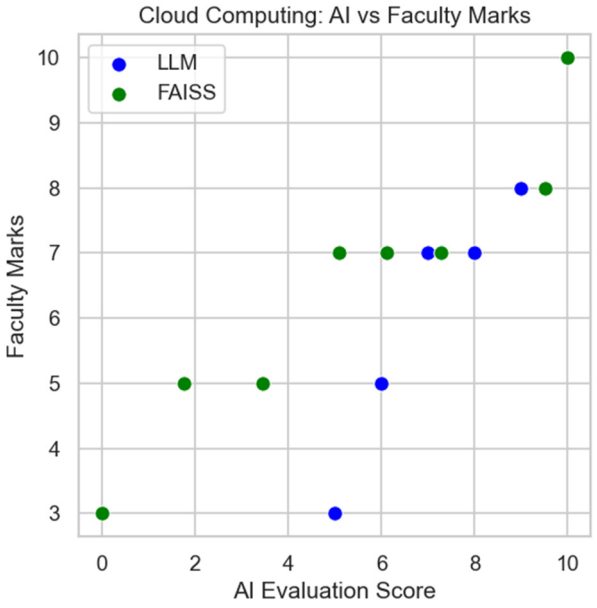


FIGURE 5. Overlaid graph of evaluation performance for Cloud Computing.

compared across **Software Engineering (SE), Knowledge Engineering (KE), and Cloud Computing**.

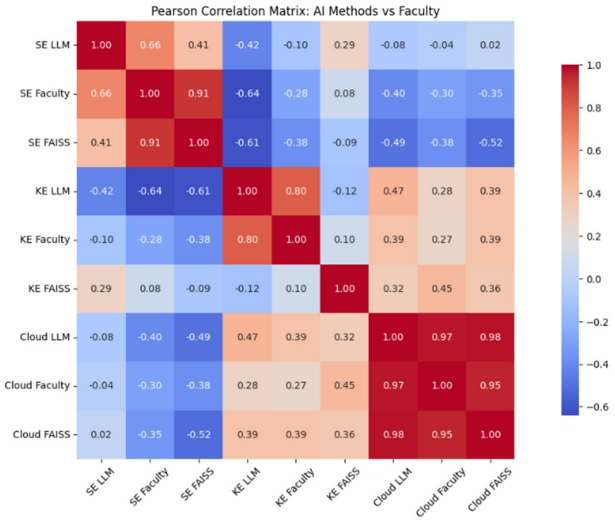


FIGURE 6. Correlation matrix of evaluation methods for SE, KE, and Cloud Computing.

1) OBSERVATIONS

i. **LLM vs. Faculty**: In Software Engineering, the LLM method shows a moderate positive correlation (0.66) with faculty marks, indicating reasonable alignment with human evaluations. However, for Knowledge Engineering and Cloud Computing, this correlation is weaker.

ii. **FAISS vs. Faculty**: The FAISS method displays a strong positive correlation with faculty marks, particularly in

Software Engineering (0.91), indicating that the FAISS model's normalized scores closely match the human evaluations.

iii. **Cross-subject correlations**: KE and Cloud have a higher degree of correlation between their respective LLM evaluations and faculty marks compared to Software Engineering. The strong correlation between Cloud LLM and Cloud Faculty (0.97) suggests that the LLM model for Cloud Computing provides highly accurate predictions when compared to the human evaluation.

Overall, the FAISS model tends to show stronger correlations with the faculty marks than the LLM model in most subjects, with the Cloud subject showing the most consistency between FAISS and faculty evaluations. The comparative study of the LLM and FAISS models highlights their respective strengths across different evaluation dimensions. FAISS exhibits a higher degree of alignment with faculty-assigned scores, particularly excelling in tasks that depend on structured, similarity-driven assessment metrics. Compared to FAISS, the LLM seems to do a better job when the answers involve more context or need a bit of reasoning. It's not just looking for facts—it actually tries to understand what the student is trying to say. FAISS, in comparison, works better when the grading is based on clear, direct similarity between what the student wrote and the model answer. So, in that sense, both have their strengths. If the evaluation mainly needs exact content matching, then FAISS might be enough. But for answers where meaning depends on how things are phrased or connected, the LLM tends to be more helpful. It probably makes more sense to pick the model depending on what kind of responses are expected in that subject or exam.

VI. DISCUSSION

This study aims to assess the efficacy of the AIvaluate system, as an LLM(**GPT-4o-mini**)–driven evaluation pipeline, with rubric scoring mode, and compares it against an embedding-based method, using Facebook AI Similarity Search (FAISS) method. The main objective is to evaluate the alignment between the scores produced by the AI compare to the scores given by faculty members across a range of subject areas.

A. INTERPRETATION OF RESULTS

The AIvaluate system's LLM-based Evaluation Agent, powered by GPT-4o-mini, showed reliable agreement with faculty grading in all 3 subjects, Software Engineering, Knowledge Engineering, and Cloud Computing. For instance, in Software Engineering, the FAISS method produced a higher Pearson Correlation Coefficient (PCC) of 0.91 over the LLM model's PCC of 0.66 representing that the FAISS method is more similar to the faculty marks. However, while FAISS has better surface-level alignment in Software Engineering, it measures based only on vector-based similarity and did not interpret anything contextually. The LLM model mediated the rubrics and provided evaluation that offered deeper semantic interpretations of student responses. In Knowledge Engineering and Cloud Computing, the LLM model provided a greater quality and deeper read on students' submissions whereas FAISS only provided basic categorization and similarity in what FAISS called 'footnotes.' This demonstrates that FAISS will typically perform well for data with mostly short answers with factual content, whereas rubric-aligned LLMs can provide more in a reasoning-based, descriptor capacity.

One recent advancement in this version of the project was based on rubric-injected prompt templates. The rubric-injected prompt templates were produced and efficiently using contextual information from the faculty-provided Answer Key, each LLM Agent was able to assess each answer as appropriate to specific question expectations. This dynamic increase in accuracy of scoring compared to the original intention of the LLM finding similarities in specific prompts.

FAISS assessment is based on relative vector semantic similarity between students and reference responses using different encoding from **all-MiniLM-L6-v2**. The FAISS implementation suggests an acceptable level of assessment which is best in questions associated to Cloud Computing, given its shorter and more technical responses. However, FAISS relies on vector representations, and therefore does not consider additional contextual information, coherence and structure of the answer. FAISS also is not aware of live rubric-based distinctions, reasoning logic, or general evaluation rules, it is a limited by fast benchmarking.

The pre-processing agent is a multi-faceted entity that incorporates the OCR cleaning agent, a spelling/alignment agent, a formatting agent, and an anonymization agent. Its overall goal is to guarantee that both models only take clean, unbiased, and evaluation-ready input. Additionally, the system includes a retry and flagging mechanism to handle LLM inference failures gracefully, further enhancing robustness and operational reliability.

B. IMPLICATIONS OF THE FINDINGS

Based on these findings, it appears that rubric-aligned LLMs are far better suited for real-world educational assessments — particularly in descriptive disciplines or long-format assessments. FAISS, while computationally optimized, is limited by its ability to respond to surface-level semantic similarity, without interpretability or feedback.

For practical use, a hybrid approach may be best, using FAISS for short, objective, low weight questions, and LLMs for subjective or long-form answers, since the system architecture accommodates this. This could help with scalability in respect to exam patterns and assessment needs.

Importantly, this rubric-injection process converts the use of an LLM from a general-purpose model to a domain-aware evaluator, since faculty prescriptive model outputs - particularly with respect to the outputs required for institutional adoption.

C. LIMITATIONS OF THE STUDY

1. There were only three subjects in the evaluation dataset with few answer scripts, thus, we are unsure how generalizable these results are across courses or grading standards.
2. Even though faculty assigned marks are used as ground truth, grading is subjective in general. And there was not comparison for inter-rater reliability.
3. The LLM is not fine-tuned on domain-specific educational sources or repositories. Though rubric injection helps alleviate this issue to a degree, performance may be better if it had been supervised fine-tuned; particularly with annotated answer-script corpora.
4. The FAISS based evaluation does not incorporate hierarchical rubric interpretation, reasoning, or contextual matching. It is only being used as part of our research benchmarking and not incorporated into the production-grade pipeline.

D. FUTURE WORK

Future plans include developing a hybrid scoring engine, that will have FAISS scoring simple keyword-based matches and your LLM scoring complex reasoning-based questions. Fine-tuning LLMs on answer-script datasets and using Bloom's Taxonomy tags in the prompts may increase grading fidelity.

More broadly, the application of AI-generated formative feedback for students will change AIvaluate from an evaluation system to a learning tool. The future development plans for the system include expanding to more subjects, languages, and writing styles for future scalability and inclusivity.

To further extend this work, future investigations will explore scalability on larger and more diverse datasets, examine inter-rater reliability in human grading as a benchmark for fairness, and develop strategies to adapt the system across varied educational contexts.

VII. CONCLUSION

The current study reports a comparison of two separate AI methods—Facebook AI Similarity Search (FAISS) and a rubric-injected LLM—to explore the feasibility of automating the grading of handwritten student answer scripts. By considering performance using the Mean Absolute Error (MAE), Root Mean Squared Error (RMSE) and the Pearson Correlation Coefficient (PCC), we provide evidence of how similar the scores produced by AI were to the assigned scores from faculty.

The findings of the study reveal that the LLM-based evaluative agent, in this case powered by GPT-4o-mini, and applied subject-based rubric-aware prompts, demonstrated consistently superior accuracy and correlation with human grading. Thus, it is particularly useful for evaluating descriptive, open-ended answers that require contextual reasoning and an understanding of meaning. Alternatively, it appears that FAISS can efficiently manage responses that are structured and technical, yet does not go beyond surface-level semantic similarity without capturing an evaluative logic from the rubric.

The architecture of the system also contains a modular, pre-processing agent that utilizes OCR-based clean-up, anonymization and improves formatting to ensure the input is clean and free of bias. Additionally, the LLM inference includes a retry and flagging process to offer a robust and reliable result from LLM inference. While FAISS was included in this study, it was used as a research benchmark and not as a production-grade scoring method.

Although the study has only examined three subjects, the designed system can be expanded to consider a larger range of academic topics. Further work will focus on hybrid scoring architectures, where FAISS is used for objective responses, and LLMs are used for a complex or subjective answer. Fine-tuning LLMS within a discipline-specific educational dataset and adding formative feedback produced by AI will enhance the capacity for automated evaluation and student learning outcomes.

Ultimately, the study offers support for the consideration of rubric-injected LLMs as an alternative approach to educational assessment, with levels of scalability, accuracy, and context. The combination of rubric inference in systems, such as AIvaluate, introduces the potential to provide automated assessment, improvements in transparency, and development of pedagogical alignments across academic disciplines.

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